2022B F=ma Exam: Problem 8

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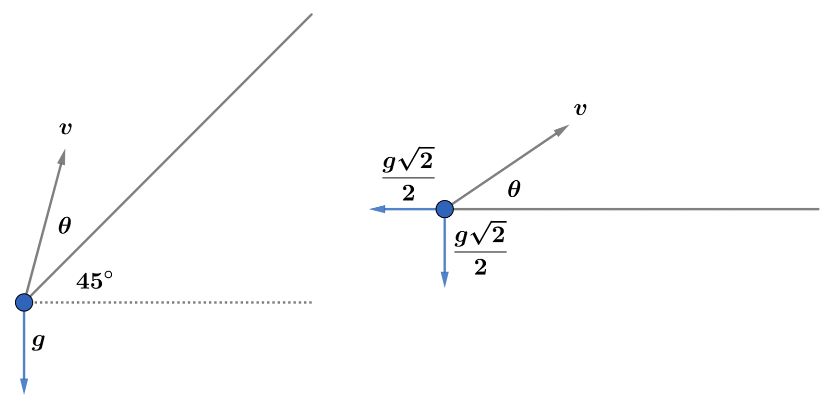

We can rotate our coordinate system such that the inclined plane becomes horizontal. The accelerations in the vertical and horizontal directions are:

$$g_y = g_x = \frac{g\sqrt{2}}{2}$$

The time it takes to go to the top is

$$v\sin\theta - g_y t = 0$$
$$t = \frac{\sqrt{2}v\sin\theta}{g}$$

One bounce is double this time:

$$T = 2t = \frac{2\sqrt{2}v\sin\theta}{g}$$

We want the displacement to be zero after 3 bounces. Our kinematics equation reads

$$0 = v_x(3T) - \frac{1}{2}g_x(3T)^2$$

Simplifying,

$$v_x = \frac{1}{2}g_x(3T)$$
$$v\cos\theta = \frac{1}{2}\left(\frac{g\sqrt{2}}{2}\right)\frac{6\sqrt{2}v\sin\theta}{g} = 3v\sin\theta$$
$$\tan\theta = \frac{1}{3},\ \theta \approx 18.4^\circ$$

so the answer is $\boxed{\text{E}}$.